Example 5.4

The following the distribution of persons according to different income groups

Income (in ₹1000)	0 – 8	8 – 16	16 – 24	24 – 32	32 – 40	40 – 48
No of persons	8	7	16	24	15	7

Find the average income of the persons.

Solution :

Direct Method:

Class	f_i	x_i	$f_i x_i$
0-8	8	4	32
8 – 16	7	12	84
16-24	16	20	320
24-32	24	28	672
32-40	15	36	540
40-48	7	44	308
Total	N =77		1956

$$\overline{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N}$$

$$= \frac{1956}{77}$$

$$=25.40$$

Short cut method:

Class	f_i	x_i		$d_i=(x_i-A)/c$	$f_i d_i$
0 – 8	8	4		-3	-24
8 – 16	7	12		-2	-14
16 – 24	16	20		-1	-16
24 – 32	24	28	A	0	0
32 – 40	15	36		1	15
40 – 48	7	44		2	14
Total	**N= 77**				**-25**

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